Technical Report: Predicting Shipment ETA & Delay Propagation

Candidate: Kalyan Dasari | **Dataset:** 300 Synthetic Shipment Journeys (Seed 42) | **Prediction Checkpoint:** Port Outer Anchorage Arrival | **Split:** 25% Stratified Test ($N=75$)

1. Problem Statement & Practical Motivation

In international maritime shipping, schedule disruptions cascade through downstream supply chain stages. A minor delay during vessel loading or ocean sailing triggers missed berthing windows, port congestion, customs clearance queues, and missed inland rail/truck connections. Traditional visibility platforms rely on static schedule tables or naive projections that assume zero future downstream delay.

This system models the **Port Arrival Checkpoint** (outer pilot station): predicting total final delay and expected arrival timestamp while the voyage is in-progress. This enables logistics dispatchers to proactively: (1) notify consignees with calibrated revised ETAs, (2) request priority berthing or off-dock container drayage to avoid yard dwell, (3) pre-lodge customs manifests (EDI), and (4) reschedule inland drayage to prevent costly port demurrage and detention charges.

> *Notice on Synthetic Data:* All data and dynamics are simulated. Performance metrics measure consistency with the underlying simulation mechanics rather than empirical real-world carrier accuracy.

2. Data Source & Synthetic Generation Assumptions

The dataset models 300 journeys across realistic trade lanes (Shanghai–Rotterdam, Singapore–Sydney, Mumbai–Dubai, Los Angeles–Hamburg, Busan–Antwerp) and container carriers (Maersk, MSC, CMA CGM, Hapag-Lloyd, ONE). The journey comprises five discrete stages:

- **Stage 1 (Departure handling):** Terminal loading and vessel dispatch.
- **Stage 2 (Ocean sailing):** Deep-sea navigational voyage.
- **Stage 3 (Port processing):** Anchorage waiting, berthing, and crane discharge.
- **Stage 4 (Customs clearance):** Regulatory document and physical inspection.
- **Stage 5 (Final delivery):** Inland drayage/rail transit to destination.

Delays propagate via linked causal equations where upstream delays compound downstream friction:

- `Dep Delay: Δdep = max(0, N(2.0 + 8.5*Weather, 4.5²))`
- `Sailing Delay: Δsail = max(0, N(3.0 + 13.5*Weather + 0.15*Δdep, 6.0²))`
- `Port Dwell Delay: Δport = max(0, N(2.0 + 26.0*Congestion + 0.22*Δsail, 7.5²))`
- `Customs Delay: Δcustoms = max(0, N(1.5 + 13.0*Congestion + 0.18*Δport, 5.0²))`
- `Final Delivery Delay: Δfinal = max(0, N(1.0 + 0.12*Δcustoms + 3.0*Weather, 4.0²))`

Congestion and weather are continuous simulated signals on a 0.0 to 1.0 scale.

3. ETA, Stage-Delay, Cumulative-Delay & Ground-Truth Definitions

To prevent accidental double-counting, the event-time model formally distinguishes **incremental stage delay** from **cumulative schedule deviation**:

- **Planned Journey Duration:** Sum of planned sailing, port dwell, customs, and final transit hours.
- **Scheduled ETA:** `Scheduled Departure + Planned Journey Duration`.
- **Incremental Stage Delay (Δs):** `Actual Stage Duration − Planned Stage Duration ≥ 0` for each stage s.
- **Cumulative Delay (Ck):** Sum of incremental delays up to milestone k. At the port checkpoint, observed upstream delay = Δdep + Δsail. At final delivery, total delay = ΣΔs.
- **Ground-Truth Actual Delivery:** `Scheduled ETA + Total Delay`.
- **Late-Delivery Label:** 1 if Total Delay > 24.0 hours; otherwise 0.
- **Predicted Final ETA:** `Scheduled ETA + Predicted Total Delay`.

4. Prediction Checkpoint & Feature Design

Predictions occur strictly at the **Port Arrival Checkpoint**. All downstream actuals are strictly withheld to eliminate target leakage. The feature pipeline enforces an explicit allowlist:

- **Schedule Baselines:** `planned_sailing_hours` (72–360h), `planned_port_hours` (8–36h), `planned_customs_hours` (4–24h), `planned_final_hours` (12–60h), `planned_remaining_hours_at_port`.
- **Observed Upstream History:** `departure_delay_hours`, `sailing_delay_hours`, `observed_delay_at_port_hours` (Δdep + Δsail).
- **Environmental Signals:** `port_congestion` (0.0–1.0), `weather_severity` (0.0–1.0).
- **Domain Interaction Features:** `upstream_delay_ratio` (observed delay / planned sailing), `congestion_weather_interaction` (congestion × weather), `congestion_buffer_ratio` (congestion / planned port hours).

Leakage Guard: Automated tests verify that forbidden downstream columns (`port_delay_hours`, `customs_delay_hours`, `final_stage_delay_hours`, `actual_port_departure`, `actual_customs_done`, `actual_delivery`, `actual_total_delay_hours`, `late_delivery`) are strictly excluded from inputs.

5. Model Approach & Evaluation Protocol

The machine learning engine pairs regression delay estimation with calibrated delay-risk classification:

- **Random Forest Regressor:** 300 decision trees, min_samples_leaf=2, max_features="sqrt", with Out-of-Bag (OOB) scoring to capture non-linear congestion interactions.
- **Delay-Risk Classifier:** Random Forest Classifier (200 trees, max depth=6) predicting calibrated probability P(Delay > 24 hours).
- **Exploratory Uncertainty Intervals:** Non-parametric Out-of-Bag residual quantiles (q10=-14.94h, q90=+16.76h).
- **Baselines:** (A) Naive Baseline: assumes zero downstream delay (Δremaining = 0); (B) Operational Heuristic: observed delay + training median downstream residual.

6. Results Table Comparing Model with Baselines

Evaluated on a 25% stratified held-out test split (N=75 shipments; seed 42) from `artifacts/eta_model.metrics.json`:

Model / Evaluation Approach	Held-Out MAE	Held-Out RMSE	Test R^2	Error Reduction vs Benchmark
Naive Baseline (Zero Future Delay)	29.168 h	32.163 h	-2.708	+66.01% error vs AI model
Operational Heuristic Baseline	11.531 h	14.175 h	0.280	Baseline Benchmark
Random Forest AI Regressor	**9.915 h**	**12.693 h**	**0.422**	**+14.01% improvement vs heuristic**

Late-Delivery Classification Performance (>24h Delay Threshold)

Metric	Measured Score	Operational Interpretation
Accuracy	**84.0%**	Correctly predicts delay status on 63 out of 75 test shipments
Precision	**87.3%**	87.3% of shipments flagged as late genuinely arrive late (low false alarms)
Recall (Sensitivity)	**95.4%**	Captures 62 out of 65 severely delayed shipments (critical for SLA protection)
F1-Score	**0.912**	Strong harmonic balance between precision and recall
ROC-AUC	**0.780**	Reliable risk discrimination across varying decision thresholds
Confusion Matrix	**[[1, 9], [3, 62]]**	TN: 1 on-time, FP: 9 false alarms, FN: 3 missed late, TP: 62 caught late

Uncertainty Interval Validation

The Out-of-Bag residual quantiles achieved **74.7% empirical coverage** on the held-out test set (against a nominal 80.0% target). Because random forest errors are heteroscedastic, these intervals are labeled exploratory rather than formal conformal guarantees.

Global Feature Importances

Top model drivers: `port_congestion` (21.42%), `observed_delay_at_port_hours` (14.26%), `congestion_weather_interaction` (13.37%), `sailing_delay_hours` (9.79%), `upstream_delay_ratio` (7.69%), `congestion_buffer_ratio` (7.53%).

7. Explainability & Worked Numerical Shipment Example

The system provides auditable explanations via exact **Tree SHAP path decomposition** ($\hat{y} = \ddot{y} + \Sigma\varphi j$) and stage-by-stage waterfalls.

Worked Numerical Example: Shipment `SHP-0001`

> **Voyage Profile:** Vessel *Nordic Horizon* (CMA CGM Group, Route: Shanghai to Hamburg via Port Delta (Dubai)).
> **Scheduled Departure:** 2025-03-15 12:00:00 UTC | **Scheduled Arrival:** 2025-04-02 00:09:36 UTC
> **Planned Stage Durations:** Sailing: 352.98h, Port Dwell: 29.31h, Customs: 19.72h, Final Inland: 18.15h (Total: 420.16h).

1. Checkpoint Observations (Port Arrival):

- Departure delay: +3.35h | Sailing delay: +17.78h (severe weather: index 0.667).
- **Observed Upstream Delay at Checkpoint:** +21.13 hours.
- **Terminal Indicators:** Port Congestion = 0.228 (favorable) | Weather Severity = 0.667.

2. Model In-Flight Predictions (Checkpoint Output):

- **Predicted Total Delay:** +41.59 hours (Estimated Downstream Remaining: +20.46 hours).
- **Predicted Final ETA:** 2025-04-03 17:45:00 UTC | **Uncertainty Band:** [26.65h, 58.35h].
- **Late Probability:** 95.9% (>24h threshold) → **Risk Status: CRITICAL**.
- **Tree SHAP Attributions:** Observed Upstream Delay (+5.19h) and Sailing Delay (+3.39h) pushed ETA later, while favorable Port Congestion (-5.96h) and generous port buffer (-3.27h) cushioned downstream delay.

3. Unknown Future Actuals (Recorded Post-Delivery for Auditing):

- Downstream Actuals: Port Dwell Delay: +11.47h, Customs Delay: +5.60h, Final Delivery Delay: +0.95h.
- **Actual Total Delay:** +39.15 hours (21.13h upstream + 18.02h downstream).
- **Ground-Truth Actual Delivery:** 2025-04-03 15:18:36 UTC.
- **Model Error:** |41.59h - 39.15h| = **2.44 hours** (The model predicted delivery within 2.5 hours of actual arrival).

8. Limitations & Real-World Deployment Requirements

8.1 Prototype Limitations

- **Synthetic Data:** Delay propagation coefficients reflect mathematical simulation assumptions rather than empirical telemetry.
- **Simplified Stationarity:** Omits seasonality (e.g., Lunar New Year, peak retail shipping), vessel blank sailings, and transshipment hub connections.
- **Discrete Checkpoint:** Evaluates only at port arrival, rather than continuous real-time updates along the ocean route.

8.2 Production Deployment Roadmap

- **Live Telemetry Ingestion:** Connect live AIS GPS satellite feeds (Spire/MarineTraffic), ocean weather forecasts (Copernicus/NOAA), and Terminal Operating System (TOS) EDI messages.
- **Rolling Temporal Validation:** Replace random splits with temporal backtesting (train on months 1..T, test on month T+1) to detect data drift.
- **Conformal Prediction:** Implement split conformal prediction to provide distribution-free, guaranteed coverage on prediction intervals.
- **Transshipment Graph Architecture:** Model multi-leg container feeder networks via Graph Neural Networks (GNNs).